

# Nauman Ahmed

*Applied AI Engineer*

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<https://portfolio-website-nauman.vercel.app/>

## Summary

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Applied AI Engineer with a software engineering background and 5+ years of combined professional and academic experience building AI-powered applications, NLP systems, agentic workflows, and scalable backend services.

Experienced in multi-agent systems, Retrieval-Augmented Generation (RAG), LLM applications, semantic search, transformer-based NLP, and workflow automation. Hands-on experience building end-to-end AI systems using LangGraph, LangChain, Hugging Face Transformers, FastAPI, PostgreSQL, pgvector, Redis, Docker, and AWS.

Strong product and engineering mindset with experience delivering 10+ web and cloud-based solutions and building production-oriented AI applications from experimentation and evaluation through deployment.

## Professional Experience

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### Full Stack Developer

Sep 2019 – Sep 2021

*StratgIT*

- Delivered 10+ web-based applications and cloud infrastructure projects for company clients, supporting end-to-end development, deployment, and production delivery.
- Built and maintained production web applications using React, Node.js, Express.js, MongoDB, and AWS.
- Developed scalable backend APIs, authentication systems, and real-time communication features.
- Designed cloud-based infrastructure using AWS services and implemented deployment workflows.
- Collaborated with cross-functional teams to deliver maintainable software solutions and improve application performance.

## Selected Projects

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### KnowledgeFlow – AI Knowledge Assistant (RAG, LangGraph, Observability)

*Next.js, FastAPI,*

*LangGraph, LangChain, PostgreSQL, pgvector, Redis, Clerk, Langfuse, Docker*

<https://github.com/nauman-ahmed/KnowledgeFlow>

- Built a production-style AI knowledge platform with streaming chat, semantic search, and citation-backed responses.
- Designed a multi-agent LangGraph workflow consisting of Planner, Researcher, and Answer Generator agents using tool calling.
- Implemented Retrieval-Augmented Generation using PostgreSQL, pgvector, embeddings, and similarity search.
- Added Redis-based memory, Langfuse observability, automated evaluations, and AI guardrails.
- Containerized services using Docker and implemented CI workflows.

### LeadFlow AI – Agentic Lead Qualification Platform (Chat, RAG, Calendar)

*Next.js, TypeScript,*

*Supabase, pgvector, RAG, Cal.com, Vapi, n8n, Docker*

<https://github.com/nauman-ahmed/LeadFlow-AI>

- Built an AI-powered lead qualification platform integrating conversational AI, appointment booking, and workflow automation.
- Implemented Retrieval-Augmented Generation using pgvector and embedding-based retrieval for context-aware customer interactions.
- Integrated Cal.com scheduling and Vapi voice-agent capabilities.
- Developed automated CRM workflows using n8n and webhook-driven integrations.
- Designed and deployed the complete full-stack architecture using Next.js, Supabase, and Docker.

### Automated Essay Scoring System

*NLP, Transformers, RoBERTa, DeBERTa, MiniLM, Optuna*

[https://github.com/nauman-ahmed/ai\\_engineering\\_lab-main](https://github.com/nauman-ahmed/ai_engineering_lab-main)

- Built an end-to-end NLP system for automated essay scoring using transformer-based models.
- Compared RoBERTa, DeBERTa, MiniLM, TF-IDF, and Word2Vec-based approaches.
- Engineered linguistic and semantic features using SpaCy, sentiment analysis, readability metrics, and TF-IDF.
- Used Optuna for hyperparameter optimization and achieved Cohen's Kappa score of 0.81.

## Education

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### M.Sc. Computer Science

2022 – Apr 2026

University of Passau, Germany

Coursework (120 ECTS): 2.4 (German Scale)

*Thesis: Multi-Agent 3D Coverage Path Planning using Liquid Neural Networks and PPO*

### B.S. Software Engineering

2016 – 2020

Usman Institute of Technology, affiliated with NED University of Engineering and Technology

## Publications

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### An Efficient Image Processing Technique to Measure and Align Vehicle Wheel Cylinder with Cloud Management System

S. Zaidi, M. Wasim, L. Ahmed, **N. Ahmed**, et al.

*KIET Journal of Computing & Information Sciences (KCIS)*, Vol. 5, Issue 1

DOI: <https://kjcisjournal.kiet.edu.pk/index.php/kjcis/article/view/90>

- Developed an automated image processing framework for cylindrical measurement and centroid alignment.
- Applied Hough Transform and Fast Finding & Fitting algorithms for circle detection.
- Achieved **95% accuracy** and **100% precision** in experimental validation.

## Technical Skills

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### Applied AI & LLM Systems

LangGraph, LangChain, AI Agents, Tool Calling, Retrieval-Augmented Generation (RAG), Semantic Search, MCP, Langfuse

### Machine Learning & NLP

Hugging Face Transformers, RoBERTa, DeBERTa, MiniLM, SpaCy, TF-IDF, Word2Vec, Optuna, Embeddings, Sentiment Analysis

### Programming

Python, TypeScript, JavaScript, SQL

### Backend & Data

FastAPI, Node.js, REST APIs, PostgreSQL, pgvector, Redis, Supabase

### Web Development

Next.js, React

### Cloud & Infrastructure

AWS(EC2, Lambda, S3, IAM), Docker, CI/CD, Linux

## Certifications

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### LangChain & ChatGPT: Production-Ready AI Applications

Feb 2026

Udemy

<https://www.udemy.com/certificate/UC-564ea2df-447a-4f38-8685-3eabd056a4c4/>

### Jenkins: Jobs, Pipelines, CI/CD and DevOps for Beginners

Jan 2025

Udemy

<https://www.udemy.com/certificate/UC-dfaf41d5-eadc-4d45-9734-6f2662821c66/>

## Languages

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English: C1

German: A2